

1. Administrative & Study Identification

Full Study Title: A Phase 2, multicenter, single-arm study to evaluate the safety and dystrophin expression after Fordadistrogene Movaparvovec (PF-06939926) administration in male participants with early stage Duchenne Muscular Dystrophy **Short Title/Acronym:** DAYLIGHT **Protocol Number:** C3391008 **NCT Number:** NCT05429372 **Sponsor Name:** Pfizer **Investigational Product:** Fordadistrogene movaparvovec (PF-06939926) - recombinant AAV9-based gene therapy **Phase of Development:** Phase II **Medical Monitor:** Dan Levy, MD, PhD (Development Head for Duchenne muscular dystrophy, Pfizer)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of fordadistrogene movaparvovec gene therapy in boys with early-stage Duchenne muscular dystrophy (DMD) at 1 year (Week 52) after dosing. **Secondary Objectives:** To assess the expression and distribution of muscle mini-dystrophin at weeks 9, 52, and year 5, and to evaluate safety outcomes through 5 years of follow-up.

2.2 Background & Rationale Duchenne muscular dystrophy (DMD) is a serious genetic disease caused by mutations in the gene encoding dystrophin, characterized by progressive muscle degeneration and weakness. This study aims to address the underlying cause of DMD in early-stage pediatric patients (before significant muscle damage develops) by delivering a gene encoding the production of a shortened but functional dystrophin protein (mini-dystrophin) into muscle cells via a one-time intravenous infusion.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 10 participants **Number of Sites:** 5 sites (located in the US and Australia) **Estimated Study Start:** 08-Aug-2022 **Estimated Primary Completion:** Late 2023 / Early 2024 (Follow-up designed for up to 5 years)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, age ≥ 2 to < 4 years old at the time of screening. 2. Confirmed genetic diagnosis of Duchenne Muscular Dystrophy (DMD). 3. Negative test for neutralizing antibodies to AAV9.

4.2 Exclusion Criteria 1. Any mutation affecting any exon between 9-13 inclusive, a deletion that affects exons 29 and 30, or a deletion affecting any exon between 56-71 inclusive. 2. Prior gene therapy treatment or any treatment designed to increase dystrophin expression within 6 months prior to screening. 3. Prior or current glucocorticoid treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of fordadistrogene movaparvovec (2E14 vg/kg). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** One-time infusion with an extensive 5-year post-treatment follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for early-stage DMD. **Prohibited:** Previous or current treatment with oral glucocorticoids or other immunosuppressive agents for the indication of DMD; exon-skipping or nonsense read-through therapies within 6 months of screening.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Pre-treatment Informed Consent, Medical History, Genetic testing, AAV9 neutralizing antibody testing, baseline labs/ECG/ECHO
2	Baseline	Day 0 Single IV Infusion of fordadistrogene movaparvovec
3	Interim	Week 9 Muscle biopsy for mini-dystrophin expression analysis
4	Primary Evaluation	Week 52 Primary safety analysis, muscle biopsy for mini-dystrophin, laboratory assessments
5	End of Study	Year 5 Final long-term safety evaluation, functional assessments, and health-related quality of life metrics

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs) through Week 52. Incidence and magnitude of abnormal laboratory findings (hematology, biochemistry, urinalysis), neurological examinations, weight, vital signs, cardiac troponin I, electrocardiogram, and echocardiography.

Measurement Method: Clinical observation, physical exams, and central/local laboratory analysis up to 1 year post-infusion.

7.2 Secondary & Exploratory Endpoints * Expression and distribution of muscle mini-dystrophin at weeks 9, 52, and year 5. * Immune responses to mini-dystrophin and the AAV9 capsid. * Change in serum creatine kinase concentration through 5 years. * Functional motor assessments and health-related quality of life outcomes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical and laboratory monitoring up to 5 years post-infusion. **Serious Adverse Events (SAEs):** Standard reporting timelines. *Note: A fatal serious adverse event occurred in a patient within this cohort in early 2024, prompting an investigation and an independent external Data Monitoring Committee (DMC) review, which temporarily paused dosing in related clinical trials.* **Data Monitoring Committee (DMC):** An independent external DMC oversees safety across the fordadistrogene movaparvovec clinical program.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled participants who received the gene therapy infusion). **Sample Size Justification:** $N=10$ is appropriate for a Phase 2, open-label, safety and biomarker expression cohort in a rare pediatric genetic disease. **Handling of Missing Data:** Standard handling for dropouts/withdrawals in longitudinal analyses up to 5 years.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Stringent safety monitoring of post-infusion immune response and hepatotoxicity/cardiotoxicity biomarkers. **Audit Trail:** eCRF logging and ongoing data clarification per standard sponsor quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: C3391008 **Registry Number:** NCT05429372 **Sponsor/Lead Organization:** Pfizer **Study Title:** DAYLIGHT **Official Regulatory Title:** A Phase 2, multicenter, single-arm study to evaluate the safety and dystrophin expression after Fordadistrogene Movaparvovec (PF-06939926) administration in male participants with early stage Duchenne Muscular Dystrophy

12. Clinical Framework (DMD Specific)

Primary Condition: Duchenne Muscular Dystrophy (DMD) **Diagnosis Threshold:** Confirmed genetic diagnosis of DMD in males aged ≥ 2 to < 4 years old **Medical Methodology:** Recombinant AAV9 vector-mediated gene therapy **Optimization Target:** Delivery and expression of functional mini-dystrophin protein in muscle cells

13. Study Procedures & Methodology

Management Pathway: Specialized infusion center management for AAV-based gene therapies **Education Protocol (HCP):** Training on proper preparation, IV infusion protocols for farnesylthioesterase 1, and acute AE monitoring **Education Protocol (Patient):** Caregiver education on post-infusion monitoring, isolation protocols (if applicable), and recognizing signs of immune response or adverse events. **Quality Audit Mechanism:** Standardized processing of muscle biopsies (Weeks 9 and 52) and central laboratory evaluation of biomarker data (e.g., troponin I, creatine kinase).

14. Observation & Follow-Up Schedule

Total Duration: 5 Follow-up years **Onsite Visit Cadence:** Baseline, Week 9, Week 52, and Annual milestones up to Year 5 **Remote Monitoring Frequency:** Regular safety follow-ups as determined by the clinical investigator. **Checklist Review Interval:** Continuous safety labs and protocol-mandated evaluations.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	Dan Levy, MD, PhD	_____	[DD-MMM-YYYY]				
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]				